

Project: Speed Dating Data

Applied Statistics and Data Analytics

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1 Background description

What influences love at first sight, or at least, attraction in the first four minutes? This dataset was compiled by Columbia Business School professors Ray Fisman and Sheena Iyengar for their paper *Gender Differences in Mate Selection: Evidence From a Speed Dating Experiment*. The data was gathered from 21 experimental speed dating events held between 2002 and 2004, in which the attendees were mostly graduate and professional students at Columbia University. During each event, every participant had a four-minute “first date” with each participant of the opposite sex. After each encounter, both individuals decided whether or not they would like to see their partner again, and rated each other on six attributes: Attractiveness, Sincerity, Intelligence, Fun, Ambition, and Shared Interests. A “match” is recorded only when both individuals expressed interest in seeing each other again. The dataset also includes pre-event, mid-event, day-after, and follow-up questionnaires capturing demographics, dating habits, lifestyle, self-perception, and beliefs about what others value in a partner.

This dataset is well suited for studying decision-making, preference formation, and gender differences in social behavior. For prediction or classification, one can model whether a given encounter results in a mutual match (*match*) or whether one participant decides “yes” on the other (*dec_o* or *dec*), using the partner’s perceived attributes, demographic similarity, shared interests, and pre-event preferences as predictors. For regression, one can model continuous outcomes such as overall liking (*like*) or estimated probability of acceptance (*prob*). The dataset also supports substantive questions:

- Do men and women weight the six attributes differently?
- How well do people’s stated preferences predict their actual decisions?
- Do participants accurately gauge how others perceive them?
- Does racial or educational similarity matter once attractiveness is controlled for?

The hierarchical structure (encounters nested within participants nested within waves) also makes the dataset suitable for discussions of multicollinearity, repeated measures, and clustered observations.

2 Data description

Speed Dating Data.csv: The data was provided from Kaggle. The file contains 8,378 records and 195 attributes. Each row corresponds to one four-minute encounter from the perspective of one participant, so each encounter appears twice in the data (once from each side), linked by participant ID and partner ID.

The six recurring attribute names throughout the dataset are: **attr** (attractive), **sinc** (sincere), **intel** (intelligent), **fun** (fun), **amb** (ambitious), and **shar** (shared interests). Many variables are repeated across multiple time points, denoted by suffixes **_1** (signup), **_s** (mid-event), **_2** (the day after), and **_3** (3–4 weeks follow-up). Variables ending in **_o** refer to the partner’s side. The main variable groups are:

- **Identifiers and event structure:** *iid, id, gender, idg, condtn, wave* (1–21), *round, position, order, partner, pid*.
- **Match outcome and pair-level features:** *match* (1 if both said yes, 0 otherwise — the main classification target), *int_corr* (correlation between participant’s and partner’s ratings across the 17 hobby items), *samerace*.
- **Partner side (**_o** variables):** *age_o, race_o; pf_o_att/sin/int/fun/amb/sha* (partner’s stated importance of each attribute); *dec_o* (partner’s decision); *attr_o, sinc_o, intel_o, fun_o, amb_o, shar_o* (partner’s ratings of you on the six attributes, 1–10); *like_o* (partner’s overall liking, 1–10); *prob_o* (partner’s estimate of probability you said yes, 1–10); *met_o* (whether partner had met you before).
- **Participant demographics and background:** *age, field* and *field_cd, undergra, mn_sat, tuition, race, imprace* and *imprelig* (importance of same race / religion when dating, 1–10), *from, zipcode, income* (median household income by zip code), *goal, date, go_out, career* and *career_c*.
- **Hobby/activity ratings (1–10):** *sports, tvsports, exercise, dining, museums, art, hiking, gaming, clubbing, reading, tv, theater, movies, concerts, music, shopping, yoga*.
- **Pre-event expectations:** *exphappy, expnum*.
- **Preference and self-perception blocks (suffix **_1**, **signup**),** six values per block, all measuring importance or rating of the same six attributes:
 - *attr1_1–shar1_1*: what *you* look for in the opposite sex.
 - *attr2_1–shar2_1*: what you think the opposite sex looks for.
 - *attr3_1–amb3_1*: your self-rating on five attributes (no *shar*).
 - *attr4_1–shar4_1*: what you think your same-sex peers look for.
 - *attr5_1–amb5_1*: how you think others perceive you.

For waves 1–5 and 10–21, the importance items are points distributed out of 100; for waves 6–9 they are 1–10 ratings. Self-rating blocks always use 1–10.

- **In-event ratings (your side):** *dec*; *attr*, *sinc*, *intel*, *fun*, *amb*, *shar* (your ratings of the partner, 1–10); *like*; *prob*; *met*.
- **Mid-event survey (suffix *_s*):** *match_es* (estimated number of matches you will have); the *_1* and *_3* preference / self-rating blocks repeated halfway through.
- **Day-after survey (suffix *_2*):** *satis_2*, *length*, *numdat_2*; the same preference and self-rating blocks repeated, plus *attr7_2–shar7_2* (your post-hoc assessment of how important each attribute actually was in your decisions).
- **Follow-up, 3–4 weeks later (suffix *_3*):** *you_call*, *them_cal*, *date_3*, *numdat_3*, *num_in_3*; the preference, self-rating, and post-hoc importance blocks repeated once more.

Table 1: Coded categorical variables.

Variable	Codes
<i>gender</i>	0 = female; 1 = male
<i>condtn</i>	1 = limited choice; 2 = extensive choice
<i>race</i> , <i>race_o</i>	1 = Black/African American; 2 = European/Caucasian-American; 3 = Latino/Hispanic American; 4 = Asian/Pacific Islander/Asian-American; 5 = Native American; 6 = Other
<i>samerace</i>	0 = different race; 1 = same race
<i>goal</i>	1 = seemed like a fun night out; 2 = to meet new people; 3 = to get a date; 4 = looking for a serious relationship; 5 = to say I did it; 6 = other
<i>date</i> , <i>go_out</i>	1 = several times a week; 2 = twice a week; 3 = once a week; 4 = twice a month; 5 = once a month; 6 = several times a year; 7 = almost never
<i>field_cd</i>	1 = Law; 2 = Math; 3 = Social Science / Psychology; 4 = Medical Science / Pharmaceuticals / Bio Tech; 5 = Engineering; 6 = English / Creative Writing / Journalism; 7 = History / Religion / Philosophy; 8 = Business / Econ / Finance; 9 = Education / Academia; 10 = Biological Sciences / Chemistry / Physics; 11 = Social Work; 12 = Undergrad / undecided; 13 = Political Science / International Affairs; 14 = Film; 15 = Fine Arts / Arts Administration; 16 = Languages; 17 = Architecture; 18 = Other
<i>career_c</i>	1 = Lawyer; 2 = Academic / Research; 3 = Psychologist; 4 = Doctor / Medicine; 5 = Engineer; 6 = Creative Arts / Entertainment; 7 = Banking / Consulting / Finance / Marketing / Business / CEO / Entrepreneur / Admin; 8 = Real Estate; 9 = International / Humanitarian Affairs; 10 = Undecided; 11 = Social Work; 12 = Speech Pathology; 13 = Politics; 14 = Pro Sports / Athletics; 15 = Other; 16 = Journalism; 17 = Architecture
<i>dec</i> , <i>dec_o</i> , <i>match</i>	1 = yes; 0 = no
<i>met</i> , <i>met_o</i> , <i>date_3</i>	1 = yes; 2 = no
<i>length</i>	1 = too little; 2 = too much; 3 = just right